

# Predicting habitat suitability, occupancy, and census size, of a rare plant using iterative adaptive niche based sampling

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## Abstract

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## 1 | INTRODUCTION

The effects of anthropogenic stressors, e.g., land use and climate change, have led to a global extinction crisis, with estimates that 20-45% of plant species are facing extinction (Brummitt *et al.* 2015; Pimm & Joppa 2015; Nic Lughadha *et al.* 2020; Bachman *et al.* 2024). Determining which plant species to focus our conservation efforts (e.g., active restoration, preserve creation, *ex situ* collections) on requires an array of data that is seldom available to decision-makers (Heywood 2017). These data generally outline simple biological and ecological parameters of species, useful for detailing their rarity and how their distributions relate to current and future anthropogenic stressors. Chief amongst these parameters are the geographic extent of occurrence (range), the distribution of suitable habitat, the occupancy of this habitat as well as the spacing of occupied patches, and the census size of individual (sub-)populations (Commission 2001; Faber-Langendoen *et al.* 2012; *USFWS species status assessment framework* 2016). While these parameters are simple, characterizing them can be time-consuming, generally requiring extensive fieldwork and travel to field sites - hence they often require proxies or heuristics for estimation in conservation assessments (Bland *et al.* 2015; Juffe-

Bignoli *et al.* 2016; Pelletier *et al.* 2018). Environmental niche models (ENMs or Species Distribution Models SDMs) have made enormous and respectable headway, respectively, in resolving the former two problems (extent, and occupancy) (Syfert *et al.* 2014; Kass *et al.* 2021; Anderson 2023), however the historic mismatch between the resolution of variables governing species distributions and the data available to serve as predictors of environmental conditions have restricted the interpretation and implementation of these models in highly heterogeneous environments (Guisan *et al.* 2013). Recent advances in remote sensing have enabled the generation of useful models in these generally biodiverse systems; further, these data offer promise for modeling additional population parameters (e.g., census size, population extent). However, the utility and use of high-resolution data have rarely been ground-verified or reported upon (Chiffard *et al.* 2020).

Recently, considerable headway has been made in generating statistically robust environmental niche models (ENMs), spurred by advances in collecting high-resolution environmental data, computing power, digitization of natural history museum records, and the acquisition of citizen science records and statistical methods, especially downsampling (Feldman *et al.* 2021; Markham *et al.* 2023). However, ENMs are rarely ground-verified, and even more seldom at landscape scales (A. Lee-Yaw *et al.* 2022). Hence, most of our knowledge about producing ENMs relies on simulated species and data, and generally at spatial resolutions considered relatively coarse to practitioners. An historic complication with the implementation and interpretation of ENM's is a mismatch between the spatial resolution of the independent variables available to model the species' fundamental niche and the factors governing the true distribution of populations - the realized niche (Lembrechts *et al.* 2019; Carscadden *et al.* 2020; Chauvier *et al.* 2022). Recent papers have had mixed results regarding the effects of imprecisely mapped occurrence data on ENM predictions, with indications that models generated in more heterogeneous environments, and at finer resolutions (e.g. ca. 3 arc-second to 10 arc-minutes ( $\sim 14.5$  km at  $38^\circ$ ) suffer minor decreases in model performance (Graham *et al.* 2008; Smith *et al.* 2023) with real species, while increasing error in mapping has drastic effects on model predictions with virtually simulated species (Gábor *et al.* 2022). A further mismatch of resolution is the year in which data on geographic localities were obtained and current conditions, which allow for positive population growth (Bracken *et al.* 2022). Historic occurrence data may now reflect conditions inhospitable to population maintenance, or even represent populations that were simply sinks from more robust populations (Bracken *et al.* 2022); using these records may decrease model performance. Collecting data on whether areas are favorable to continued recruitment from the soil seed bank is perhaps more pertinent than assessing whether long-lived individuals persist.

ENM models generally suffer from having few, often spatially biased, occurrence records as dependent

variables, which often fail to characterize the ecological breadth of the species (Feeley & Silman 2011; Stolar & Nielsen 2015). Accordingly, while many ENMs achieve high performance on small hold-out subsets, they are unlikely to detect many new populations during ground verification (A. Lee-Yaw *et al.* 2022). To increase the number of presences and absences in sites that are relatively similar to those harboring presences, which can be used for training models, iterative adaptive-niche-based sampling (ANBS) has become increasingly employed (Guisan *et al.* 2006). In ANBS cells with a high probability of occurrence or where models have higher prediction uncertainty, are preferentially visited. After each round of field visits, a new model is fit, incorporating both the original and the most recent data (Mondain-Monval *et al.* 2024). Using this process not only allows for the acquisition of a larger number of presences and absences, but also allows for verifying coordinate placement, confirming that historic points are still extant, reducing model uncertainty, and acquiring additional data such as census estimates and life stages (Stockwell & Peterson 2002; Wisz *et al.* 2008; Jansen *et al.* 2022). However, few studies have reported field-verified ANBS sampling efforts of plant species (Borokini *et al.* 2023; Johnson *et al.* 2023; Zimmer *et al.* 2023).

An ENM predicts a single variable: the probability of suitable habitat for the taxon (taddeo2024grimes). Although in most conservation applications, the insight desired from an ENM is the species realized distribution (cite). The bridge between an ENM and a plant population’s presence is related to the dispersal of propagules and the establishment of the population, rather than the distribution of the fundamental niche alone (Engler & Guisan 2009). Calls - and significant progress have been made for better integration of SDM predictions with models of dispersal since the rise in popularity of ENMs (Guisan & Thuiller 2005; Franklin 2010). Most of this progress focuses on the goal of ‘connecting’ currently suitable habitats to habitats predicted to be suitable under climate change scenarios (Engler & Guisan 2009; Bocedi *et al.* 2014). Accordingly, ground verification has not been conducted for them, nor has apparently much effort been made to improve ground verification for rare species... To assist practitioners in detecting new populations or extending the range of known populations, we propose modeling plant occupancy as a random variable that depends on distance from known occupied sites, the size and shape of the target site, and its relation to other suitable habitats. In simple terms, these perspectives link habitat suitability to the tenets of island biogeography (CITE). ‘Distance’ may be defined as Euclidean (or Haversine for large distances), or as a least-cost distance reflecting a generalized surface which conveys the difficulty for seeds to travel between the nearest occupied sites and the site of interest (Etherington 2016). Landscape metrics postulated to relate to site occupancy include the patch metric Core Area Index, roughly reflecting the probability of a propagule arriving at a larger patch, and the class metrics proximity and contagion, both of which reflect aggregation among occupied sites.

Obtaining reliable estimates of plant population census sizes can be a time-intensive process requiring two pieces of information: 1) measurements of plant density and 2) population extent. Instead of directly determining either the population’s boundaries or density, observers estimate population sizes; however, these estimates are often unreliable (Reisch *et al.* 2018). Several promising field methods for acquiring density measurements have recently been developed (Schorr 2013; Alfaro-Saiz *et al.* 2019; Rominger & Meyer 2019; Ermakova *et al.* 2021; Krening *et al.* 2021) to estimate better effective population census size ( $N$ ), as well as progress in the application of genomic methods to estimate effective population sizes ( $N_e$ ) (e.g. linkage disequilibrium) (Waples 2024). However, these methods are focused on descriptive rather than predictive processes, estimating uncertainty in measurements *within* a population rather than across the species’ range. We propose that estimates of plant density are best generated using methods that allow fitting statistical models incorporating spatial covariates and that predict estimates and measurements of uncertainty across gridded surfaces (Oliver *et al.* 2012; A. Lee-Yaw *et al.* 2022; Doser *et al.* 2024).

[Figure 1 about here.]

Mapping the boundaries of a population is another time-consuming, albeit essential, task for realistic estimates of population census size. Currently, population boundaries are implicitly delineated by practitioners walking distances beyond which gene flow between individuals should be minimal (e.g., 1km), searching for additional individuals, or via aerial imagery (Elzinga *et al.* 1998). Many rare plant species are small and inconspicuous, and many taxa grow in small, clustered groups in specific habitats, making detection via either method difficult (Condit *et al.* 2000; Chen *et al.* 2013). High-resolution spatial data eventually enable the determination of the extent of individual populations by detecting the edges of suitable habitat. Here, we 1) assess the status of historic occurrence records, 2) develop ENMs at three spatial resolutions; 3) estimate population census size using 3a) models of plant density 3b) and delineated population boundaries; 4) use connectivity analysis in an ANBS context to conduct field work to identify new occurrences. Collectively, these steps all build from an ENM, and showcase its utility in serving as a variate and derived product for multiple facets of describing

## 2 | METHODS

### Study Species & System

*Eriogonum coloradense* Small (Polygonaceae) is a synoecious mat-forming perennial herb endemic to the Central Southern Rocky Mountains in Colorado, U.S.A. It's known geographic range covers five counties, with 22 formally described populations, and it is thought to occur across a range of elevation, slopes, aspects, soil types, and habitats. Its elevation range spans 2703 - 3914 m, and the major habitat types it's known from include high-elevation sagebrush steppe, subalpine grasslands, and alpine slopes. It is considered an S3/G3 species by NatureServe and a Tier 2 species in the State Wildlife Action Plan by the Colorado Parks and Wildlife Service (Anderson 2004).

### Data Acquisition

Dependent data were gathered from iNaturalist (80 records) and the Consortium of Southern Rockies herbaria (131 records) ['Southern rocky mountain herbaria portal'; INATURALIST]. These data were manually reviewed, and 16 herbarium records with low geolocation quality or that were georeferenced to localities that did not match their herbarium labels were removed.

Digital Elevation Models at 3 arc (type) and 1 arc (type), and Digital Elevation Products at 1/3 arc and 1m resolution were acquired from the United States Geological Survey and clipped to the analysis domain (a rectangle buffered 16km (10 mi) beyond any known population). A 3m-resolution digital elevation model, which is not available at native resolution from the USGS for this area, was created by bilinear resampling of the 1m and 10m data. 1m-resolution data were available for roughly XX% of the domain (which contained %% of known populations); the remaining species domains' 3m data were created by linear resampling of the 10m DEP. A comparison of these results in the overlapping region is presented in Appendix XX. These elevation products were used to create all geomorphology data sets using whiteboxTools (Wu & Brown 2022). Vegetation cover data were generated by combining raster data into continuous cover classes: Forested, Shrub, and Herbaceous (Tuanmu & Jetz 2014).

ClimateNA was used to create a dataset at 3 arc-second resolution, which was then bilinearly interpolated to generate products at finer resolutions (Wang *et al.* 2016). Gray co-occurrence level matrices were produced with the glcm R package from 2023 NAIP aerial imagery, bilinearly resampled to the original resolution, using default settings with a window size of 5 in both directions (Zvoleff 2020).

## Ground Verification

The first round of ground verification was carried out from June to September 2024. All pre-existing occurrence points were considered candidates for revisits, and 25 trails leading to them were marked as sample areas. Each trail was manually mapped and buffered by 45m in each direction; 551 random points were drawn, thinned to distances  $> 90$  m, leaving 309 random plots for assessment. 13 trails were visited, allowing for the assessment of 171 random points and 20 new occurrence records. When conducting field work, all presences of *E. coloradense* were opportunistically noted, and to better describe the spread of the population points were subjectively placed, with considerations of local abundance, ca. 30-50m from the previous one until traversing out of the population ( $n = 246$ ). Additionally, subjectively placed absences were also collected in areas that seemed favorable, or were in proximity to *E. coloradense* individuals; this occurred both in the field ( $n = 129$ ) and through the use of aerial imagery on a computer afterward ( $n = 66$ ).

Adaptive Niche-Based Sampling was conducted in July 2025. To determine whether adaptive sampling performed better than alternative sampling regimes (e.g., stratified, random), and if it could be improved by occupancy modeling, 30 points were selected for each of the aforementioned sampling schemas, plus 30 occupancy points.

## Comparison of Different Spatial Resolutions

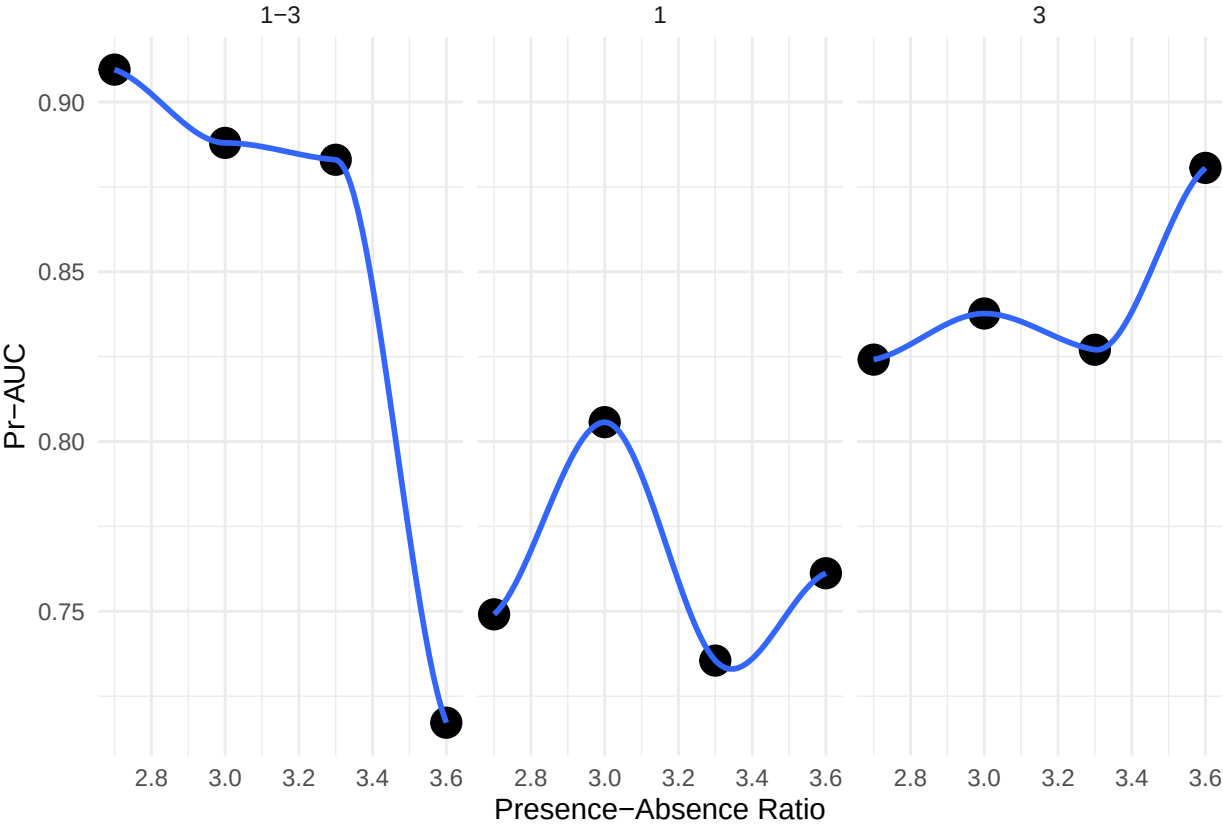
Environmental Niche Models were generated at three spatial resolutions, 3 (~72m), 1 (~24m), and 1/3 (~8m) arc-seconds. The modeling details were similar across resolutions.

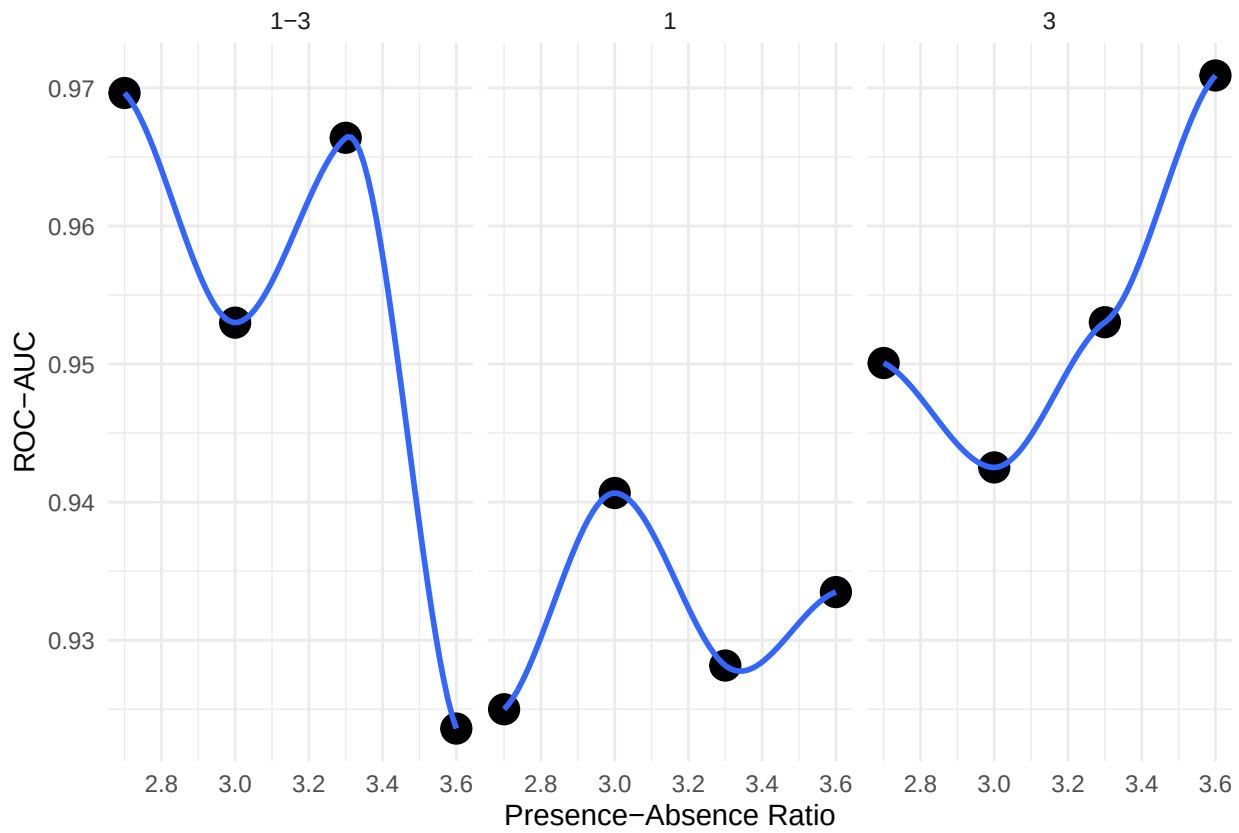
Records were thinned to the distance of a hypotenuse of a cell to avoid pseudo-replicates (Aiello-Lammens *et al.* 2015). 1.1 times as many Absence, as presence, records were generated using the background function with method environmental distance from sdm (Naimi & Araujo 2016). These records were then manually reviewed, and six records deemed to be in areas with possible presences were identified and removed. After this, the records were randomly sampled to reduce the data set to XX records. After the first modeling iteration, all additional presences and absences were thinned similarly and combined with the original absence records. ‘Presences’ with greater coordinate uncertainty than the modeling resolution were removed.

All random forest models were fit using the ranger package, 95% confidence intervals were generated using the infinitesimal jackknife (Wager *et al.* 2014; Wright & Ziegler 2017).

**Adaptive and Occupancy Based Field Sampling**

500 stratified points ranging from 1-100% probability of suitable habitat were generated using sample (terra). Occupancy scores for each of these points were then calculated using (whiteboxtools). 200 points which maximized the spread of values along both dimensions (habitat, occupancy) were then chosen for ground verification.

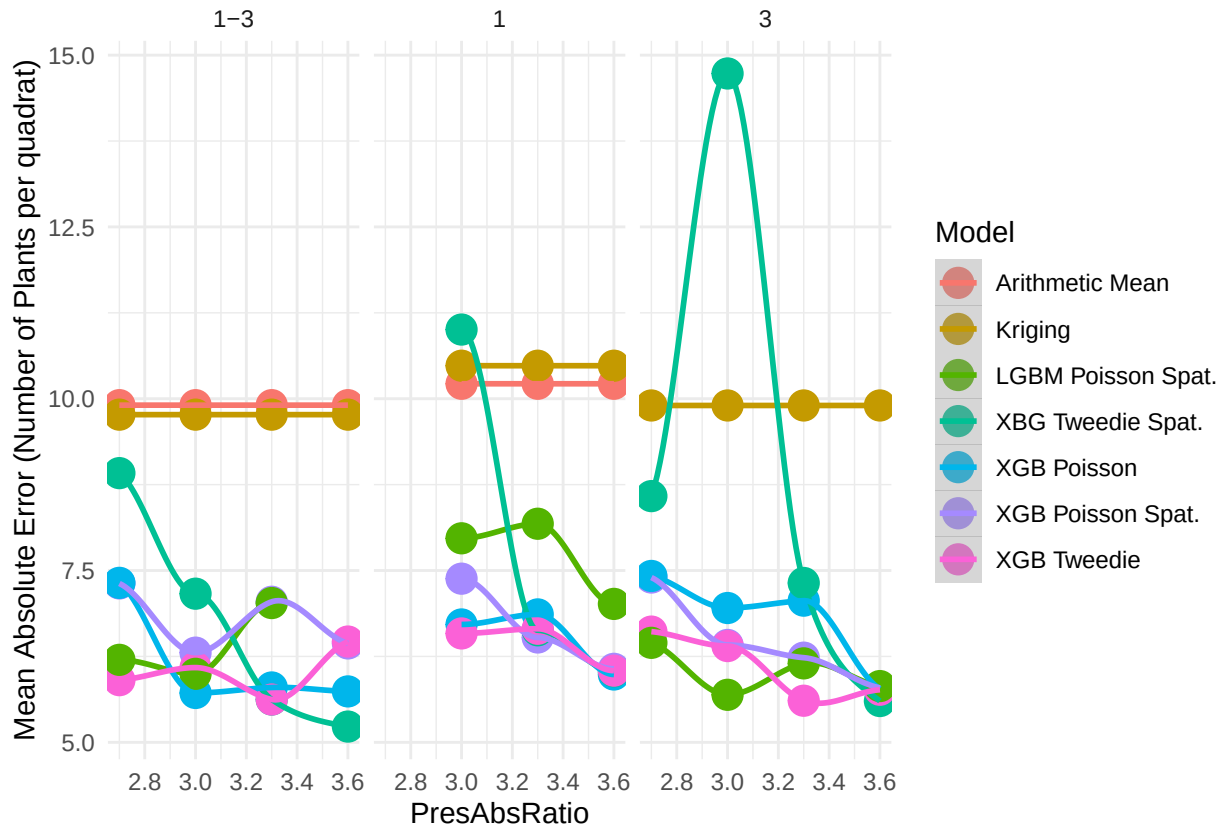




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#### 174 **Plant Density**

175 Plant density was modelled using five methods, focusing on a comparison of spatial and non-spatial cross-  
 176 validation methods.



## Species Occupancy

## 3 | RESULTS

### Comparison of Different Spatial Resolutions

The area under the receiver operating curve (AUC-ROC) and precision-recall curve (AUC-PR), the True Skill Statistic, and Sensitivity and Specificity (Allouche *et al.* 2006; Sofaer *et al.* 2019).

## Ground Verification

## Plant Density

## Species Occupancy

## Simulations of Sample Size

## 4 | DISCUSSION

## 5 | CONCLUSIONS

## 6 | ACKNOWLEDGMENTS

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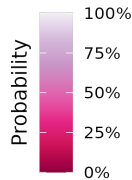
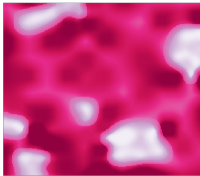
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<sup>343</sup> **List of Figures**

<sup>344</sup>     1     figgy stardust . . . . . 16

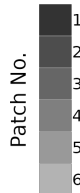
# Relationships between facets of ENMs

## Suitable Habitat



The probability of suitable habitat indicates areas which range from favorable (100%) to inhospitable (0%) for the species, based on the training data. Different techniques are used to interpret this probability, and to apply thresholding to it... This probability can then be used as a covariate in further modelling (e.g. density), or derived for other applications such as generating patch shapes and sizes, noting occupancy, and the distance between occupied and unoccupied patches.

## Suitable Patches

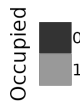


Species are limited to areas of habitat which are large enough to support positive population growth, or which serve as sinks of propagules.

Given the density of populations, and size of individual plants, not all patches have enough area to support an effective population.

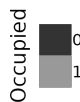
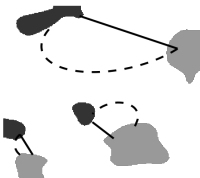
Further, species vary in their ability to grow on peripheral or marginal habitat around optimal habitat, decreasing the true patch size. Finally smaller patches create smaller "targets" for propagules to disperse to, decrease the probability of plant establishment.

## Patch Occupancy



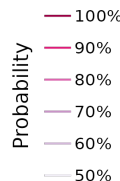
Training data explicitly denote the occupation status of certain areas of suitable habitat. Using these data we can mark patches of suitable habitat which contain known individuals or populations. These occupied patches can be combined with measures of distance to set up an adaptive sampling schema to find new populations.

## Distances



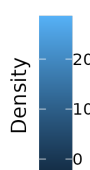
Patches are separated by both geographic (measured as euclidean for short distances or Haversine to accomodate the curvature of the earth) and environmental distances. These distances decrease the likelihood of propagules dispersing between patches. Environmental distances can be estimated by first creating generalized least-cost surfaces and then using various cost algorithms (e.g. electrical conductivity) to link patches known to be occupied with unvisited patches of suitable habitat.

## Boundaries



Different thresholds of habitat probability can be used with field verification to most accurately determine where populations end. These boundaries can then be used to refine patch shapes, the distances between them, and to estimate census sizes of the populations in a final modelling process.

## Plant Density



The number (abundance) of plants per specific area can be modelled and projected onto gridded surfaces. This modelling process can involve using the ENM probability surface as a covariate and clipping estimates to predefined boundaries. Summing values from all raster cells across cells can produce an estimate of the number of plants per (sub-)population.

Figure 1: figgy stardust